

<limits.h>

```
#define CHAR_BIT <#if expression >= 8>
#define CHAR_MAX <#if expression >= 127>
#define CHAR_MIN <#if expression <= 0>
#define INT_MAX <#if expression >= 32,767>
#define INT_MIN <#if expression <= -32,767>
#define LONG_MAX <#if expression >= 2,147,483,647>
#define LONG_MIN <#if expression <= -2,147,483,647>
#define MB_LEN_MAX <#if expression >= 1>
#define SCHAR_MAX <#if expression >= 127>
#define SCHAR_MIN <#if expression <= -127>
#define SHRT_MAX <#if expression >= 32,767>
#define SHRT_MIN <#if expression <= -32,767>
#define UCHAR_MAX <#if expression >= 255>
#define UINT_MAX <#if expression >= 65,535>
#define ULONG_MAX <#if expression >= 4,294,967,295>
#define USHRT_MAX <#if expression >= 65,535>
```

Include the standard header **<limits.h>** to determine various properties of the integer type representations. The standard header **<limits.h>** is available even in a [freestanding implementation](#).

You can test the values of all these macros in an [if directive](#). (The macros are [#if expressions](#).)

CHAR_BIT

```
#define CHAR_BIT <#if expression >= 8>
```

The macro yields the maximum value for the number of bits used to represent an object of type *char*.

CHAR_MAX

```
#define CHAR_MAX <#if expression >= 127>
```

The macro yields the maximum value for type *char*. Its value is:

- [SCHAR_MAX](#) if *char* represents negative values
- [UCHAR_MAX](#) otherwise

CHAR_MIN

```
#define CHAR_MIN <#if expression <= 0>
```

The macro yields the minimum value for type *char*. Its value is:

- [SCHAR_MIN](#) if *char* represents negative values
- zero otherwise

INT_MAX

```
#define INT_MAX <#if expression >= 32,767>
```

The macro yields the maximum value for type *int*.

INT_MIN

```
#define INT_MIN <#if expression <= -32,767>
```

The macro yields the minimum value for type *int*.

LONG_MAX

```
#define LONG_MAX <#if expression >= 2,147,483,647>
```

The macro yields the maximum value for type *long*.

LONG_MIN

```
#define LONG_MIN <#if expression <= -2,147,483,647>
```

The macro yields the minimum value for type *long*.

MB_LEN_MAX

```
#define MB_LEN_MAX <#if expression >= 1>
```

The macro yields the maximum number of characters that constitute a [multibyte character](#) in any supported [locale](#). Its value is $\geq$ [MB_CUR_MAX](#).

SCHAR_MAX

```
#define SCHAR_MAX <#if expression >= 127>
```

The macro yields the maximum value for type *signed char*.

SCHAR_MIN

```
#define SCHAR_MIN <#if expression <= -127>
```

The macro yields the minimum value for type *signed char*.

SHRT_MAX

```
#define SHRT_MAX <#if expression >= 32,767>
```

The macro yields the maximum value for type *short*.

SHRT_MIN

```
#define SHRT_MIN <#if expression <= -32,767>
```

The macro yields the minimum value for type *short*.

UCHAR_MAX

```
#define UCHAR_MAX <#if expression >= 255>
```

The macro yields the maximum value for type *unsigned char*.

UINT_MAX

```
#define UINT_MAX <#if expression >= 65,535>
```

The macro yields the maximum value for type *unsigned int*.

ULONG_MAX

```
#define ULONG_MAX <#if expression >= 4,294,967,295>
```

The macro yields the maximum value for type *unsigned long*.

USHRT_MAX

```
#define USHRT_MAX <#if expression >= 65,535>
```

The macro yields the maximum value for type *unsigned short*.

See also the [Table of Contents](#) and the [Index](#).

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